

OLG SOFTWARE ENGINEERING PREPARATION PLAN

6-Month timetable for OLG first, while building a competitive Canadian software-engineering profile

Target roles: Junior Software Developer • Software Developer • QA/Software Testing • Application Support • Technology Analyst

Core stack: Python • C++ • JavaScript • SQL • FastAPI • React • PostgreSQL • Git/GitHub • Testing • Docker • AWS

Weekly commitment: 18–22 focused hours. Start applying from Month 3; do not wait until the roadmap is finished.

Weekly Study Schedule

Day	Focus	Time	Output
Monday	Python / backend fundamentals	2.5–3 h	Lesson + coding exercises
Tuesday	SQL + PostgreSQL	2.5–3 h	10–15 SQL problems
Wednesday	DSA in C++/Python	2.5–3 h	2–3 quality problems
Thursday	FastAPI / React	2.5–3 h	Feature implementation
Friday	Testing + Git + engineering	2–2.5 h	Tests / PR / debugging
Saturday	Project build + interview prep	4–5 h	Major project milestone
Sunday	Review + mock interview + applications	2–3 h	Weekly review + applications

Month-by-Month Roadmap

Month	What to learn/build	Exit target
Month 1 — Foundations	Python: functions, data structures, OOP, exceptions, modules, typing, async basics. SQL: SELECT, JOIN, GROUP BY, CASE, subqueries. DSA: arrays, strings, hashing, two pointers, sliding window. Git basics.	50 DSA problems • 50+ SQL exercises • 1 small Python project • GitHub profile cleaned up
Month 2 — Backend	FastAPI: routing, Pydantic, validation, dependency injection, REST, authentication/JWT, authorization, error handling. PostgreSQL: relationships, indexes, transactions, schema design. Postman + pytest.	Working FastAPI + PostgreSQL API • Authentication • 20+ API tests • API documentation
Month 3 — Flagship Project	Rebuild Driveway Docs with React + FastAPI + PostgreSQL. Implement customer/admin roles, booking CRUD, validation, authentication, search/filtering, error handling and tests.	Deployable flagship project • Professional README • Architecture diagram • Start OLG + Canadian applications
Month 4 — Professional Engineering	Docker and Docker Compose. CI basics. Testing deeper: unit/integration/API tests. Logging and configuration. React integration and clean UI. Continue DSA and SQL.	Dockerized application • CI pipeline • Strong test suite • 100–120 total DSA problems
Month 5 — Enterprise Readiness	AWS fundamentals: EC2, S3, RDS, IAM, CloudWatch. System design: APIs, caching, queues, scaling, database indexing, rate limiting, monitoring. Build a second enterprise-style project.	Second project • Cloud deployment • 3 system-design exercises • OLG technical interview preparation
Month 6 — Interview & OLG Push	Daily interview practice. DSA medium problems, SQL timed questions, project deep-dives, system design, STAR behavioural stories and OLG research. Apply broadly while prioritizing OLG.	120–150 quality DSA problems • 50+ SQL problems • 10 mock interviews • 30+ targeted applications

OLG Interview Preparation

Technical interview

Be able to code without tutorials, explain Big-O, write SQL joins/window functions, design REST APIs, explain JWT authentication, discuss PostgreSQL indexes/transactions, and troubleshoot API failures.

Project interview

Know Driveway Docs deeply: requirements, architecture, database schema, API flow, authentication, testing, trade-offs, bugs you solved, and what you would change at 10x scale.

Behavioural interview

Prepare STAR stories for teamwork, conflict, failure, feedback, learning quickly, initiative, customer focus, pressure, prioritization and adapting to change. Connect answers naturally to OLG's workplace culture.

Application strategy

From Month 3 onward, apply to suitable OLG technology roles plus Workiva, Pollard Banknote, local Sault employers, banks, fintechs, consulting companies and remote Canadian software roles. Tailor the resume to each posting.

Daily Minimums

Area	Time	Rule
DSA	30–45 min	2–3 problems on study days; explain approach and complexity.
SQL	30–45 min	Practice queries and database reasoning.
Project	60–90 min	Build, test, refactor, document — do not just watch tutorials.
Interview	20–30 min	Explain one technical topic aloud or practice one STAR story.

Non-Negotiable Milestones

- By end of Month 1: comfortable with Python fundamentals, SQL basics and core DSA patterns.
- By end of Month 2: independently build and test a FastAPI + PostgreSQL REST API.
- By end of Month 3: Driveway Docs is a genuine full-stack application and you begin applying.
- By end of Month 4: project is tested, Dockerized and professionally documented.
- By end of Month 5: understand AWS basics and can discuss basic system design.
- By end of Month 6: interview-ready for entry-level/junior software, QA and application-support technology roles.

Important Rule

Do not wait six months before applying. The objective is to become employable while learning. If an appropriate OLG posting appears in Month 2 or 3, apply. Continue improving while interviewing.

This roadmap is designed to make OLG the first target while keeping your profile competitive for other Canadian technology employers.